EE332 Introduction to VLSI—Midterm

1. (40 points) Answer the following problems.
   (a) What is antenna effect? (4 points)
   (b) What is channel length modulation effect? (4 points)
   (c) Draw a CMOS transistor-level schematic realizing the function $f = \overline{xy + (u+v)w}$. (6 points)
   (d) List four properties of static CMOS logic. (4 points)
   (e) List two latchup prevention approaches. (4 points)
   (f) Draw a CMOS inverter and label the drain and source of the PMOS and NMOS transistors. (4 points)
   (g) Define the low and high noise margins using the voltage transfer curve of a CMOS inverter. (4 points)
   (h) List two possible approaches for alleviating the effect of crosstalk on two neighboring parallel interconnects (4 points).
   (i) List the three major components of the load capacitance of a CMOS logic gate (6 points).

2. (20 points) Figure 1(a) depicts a CMOS logic circuit. Assume that the delay of the inverter is 1 delay unit, where the inverter size is $W_P/L_p = 2\mu m/0.18\mu m$ and $W_n/L_n = 1\mu m/0.18\mu m$, and the input capacitance of the inverter is C_i. Also, assume that the size of each NMOS transistor and PMOS transistor of all the other gates is the same as that of the inverter.
   (a) (10 points) Determine the delay of each gate in terms of delay unit (neglect the diffusion and wire capacitance).
   (b) (10 points) Assume that a dual-V_t process can be used to implement the CMOS circuit. Draw the transistor schematic diagram of the CMOS circuit in terms of the symbols shown in Fig. 1(b) such that the leakage power of the CMOS circuit is minimal without sacrificing the speed of the CMOS circuit.

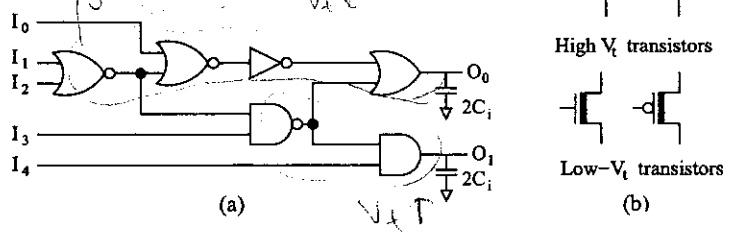


Figure 1: (a) A CMOS circuit. (b) Symbols of high-V_t and low-V_t transistors.

3. (20 points) Consider a buffer chain shown in Fig. 2, which is used to drive a large capacitance C_L. Assume that the buffer size of the first stage is the minimum size which input capacitance is C_i. Also, each stage in the buffer chain should have its transistor width larger than the previous one by a factor β. Neglect the effect of parasitic capacitance.
   (a) (10 points) How much are the values of the size ratio β and the number of stages N such that the delay of the buffer chain is minimal?

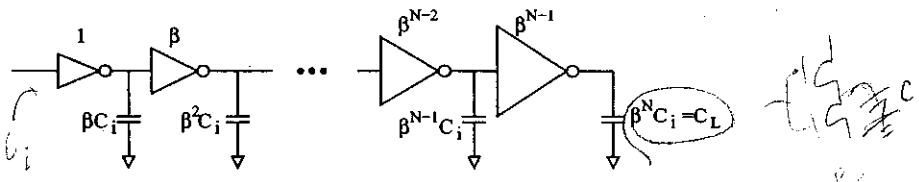


Figure 2: A tapper buffer for driving a large capacitance.

(b) (10 points) Calculate the power efficiency η ($\eta = \frac{P_T - P_L}{P_T}$) when the optimized ratio is used to design the buffer chain. Here, P_T and P_L denote total switching power of the buffer chain and the load capacitance C_L, respectively.

4. (10 points) The figure shown in Fig. 3 depicts an inverter chain for driving a capacitive load C_L. Assume that every inverter has a minimum-size NMOS and the ratio of PMOS to NMOS is α. Let $\mu_n = 2\mu_p$ and the resistance and input capacitance of the minimum-size NMOS are R and C, respectively. Derive the value of α such that the T=6RC and the power consumption constituted by the first two inverters is minimal.

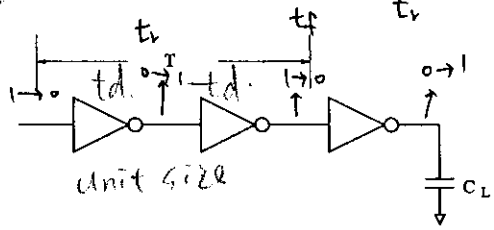


Figure 3: A inverter pair.

5. (10%) Table 1 shows the truth table of a binary full adder, where c_i, s, and c_o denote the carry input, sum output, and carry out. Design a full adder using CMOS logic with PMOS and NMOS transistors.

Table 1: Truth table of a binary full adder.

x	y	c_i	s	c_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1